

Estimator mathematics

B01 — giving the battery terminals, so there is something to measure rather than a state to read.

Symbols

z	state of charge, in $[0, 1]$	–
$OCV(z)$	open-circuit voltage at that charge	V
I	pack current, POSITIVE discharging	A
R	internal resistance	Ω
V	terminal voltage — what a voltmeter reads	V
Q	usable charge capacity	Ah
η	coulombic efficiency	–
b_I	current-sensor BIAS — the term that matters	A

1. The plant

A battery hides its charge. What it exposes is a voltage, pulled down by whatever current is being drawn through its own resistance:

$$V = OCV(z) - IR$$

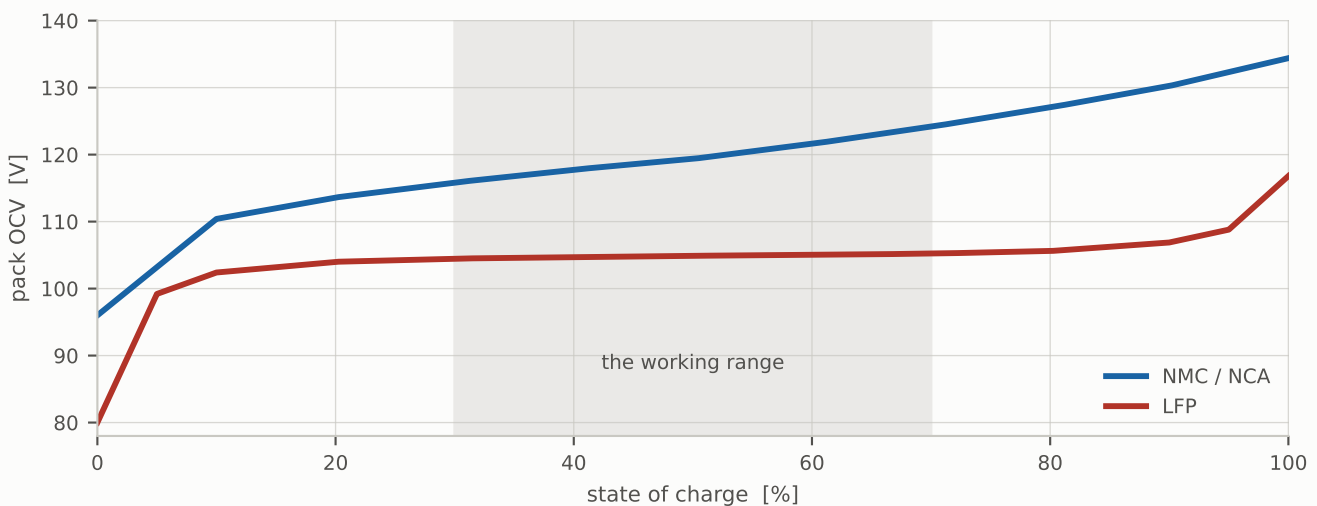
That single term is what makes estimation hard rather than arithmetic. A heavy discharge LOOKS like a low state of charge, and separating the two is the filter's entire job.

Charge itself moves by integration — coulomb counting:

$$\frac{dz}{dt} = -\frac{\eta I}{3600Q} \quad \Rightarrow \quad z_{k+1} = z_k - \frac{\eta I_k \Delta t}{3600Q}$$

Δt in seconds, Q in amp-hours, hence the 3600.

Pack OCV for the two candidate chemistries, 32 cells in series. The shaded band is where the outpost actually lives.



2. Choosing the OCV curve

Five ways to write $OCV(z)$, and why the least clever one wins here.

Analytic forms, in rough order of how often they appear in the literature:

Polynomial

fits anything; monotonicity not guaranteed

$$OCV(z) = \sum_{i=0}^n a_i z^i$$

Shepherd

diverges as $z \rightarrow 0$

$$OCV(z) = E_0 - \frac{K}{z} + Ae^{-Bz}$$

Nernst

diverges at BOTH ends

$$OCV(z) = E_0 + K_1 \ln z + K_2 \ln(1 - z)$$

Plett combined

accurate mid-range; same endpoint problem

$$OCV(z) = K_0 - \frac{K_1}{z} - K_2 z + K_3 \ln z + K_4 \ln(1 - z)$$

Lookup + linear

monotonic by construction

$OCV(z)$ interpolated between breakpoints

Take the lookup table

Not because it is simpler, but because three of its properties are load-bearing here:

- **MONOTONIC BY CONSTRUCTION.** Check the breakpoints once and the curve can never fold back on itself. A fitted polynomial can, and where it does, SoC becomes ambiguous — two charges, one voltage.
- **NO ENDPOINT DIVERGENCE.** Every analytic form above except the polynomial carries a $\ln z$ or a $1/z$. This pack's floor is $z = 0.05$, close enough to the singularity that the fit is doing its worst work exactly where the outpost spends 554 hours of every night.
- **IT PORTS.** A const array and two multiplies cross-compile to an ESP32 without a $\ln$ in sight, and swapping NMC for LFP is swapping a table.

Whichever form you take, one property is not optional:

$$\frac{dOCV}{dz} > 0 \quad \text{everywhere}$$

Page 4 is about what happens when it is merely small.

3. Why a filter at all

Two ways to know the charge. Both wrong, and wrong differently.

With perfect instruments the terminal equation inverts and no filter is needed:

$$z = \text{OCV}^{-1}(V + IR)$$

Instruments are not perfect, and the two available methods fail in opposite directions — which is precisely what makes them worth fusing.

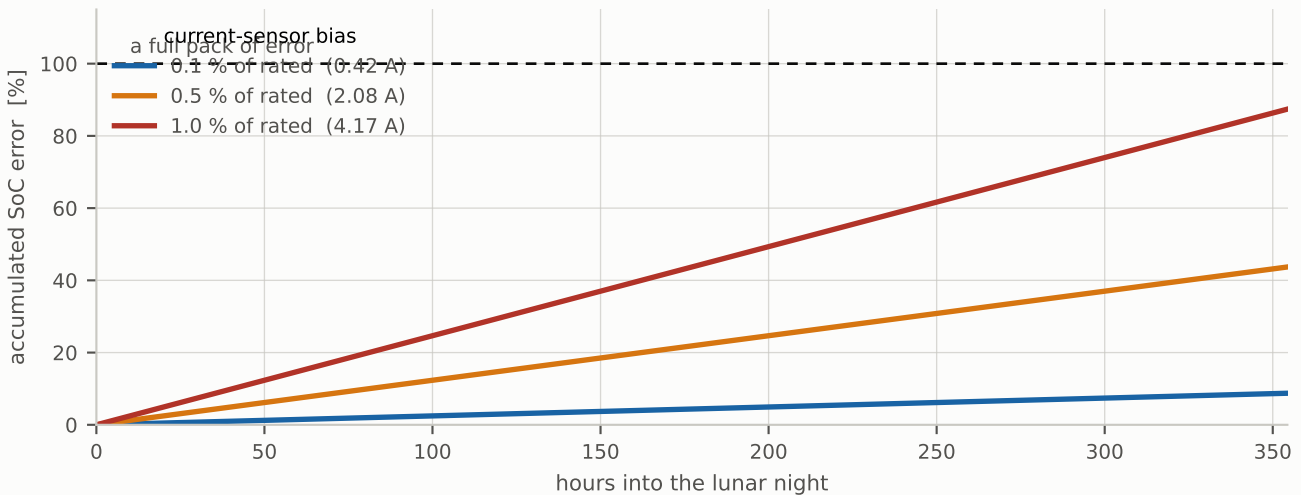
Coulomb counting — precise, and biased

Integrating current is exact over short intervals. But a sensor BIAS integrates too, and never washes out:

$$\varepsilon_z(t) = \frac{b_I t}{3600 Q}$$

With $Q = 1689$ Ah and a bias of just 0.5 % of the 417 A rating:

Drift across one lunar night. Nothing about this is recoverable from current alone — the error is a straight line with no



A 0.5 % bias reaches 44 % SoC error by dawn. The controller would shed life support on a pack that was nearly full, or fail to shed on one that was nearly empty.

Voltage inversion — unbiased, and blunt

Reading charge from voltage has no memory, so no drift. Its error is instead whatever error sits in V and IR , divided by the curve's slope:

$$\varepsilon_z = \frac{\Delta V + I \Delta R}{d\text{OCV}/dz}$$

An EKF runs coulomb counting as its PREDICT step and voltage as its CORRECT step, so the drift is continually anchored by a measurement that cannot drift.

4. Observability

The slope of the curve is not a detail. It is the whole question.

The measurement Jacobian — how much the predicted voltage moves when the estimated charge moves — is just the curve's slope:

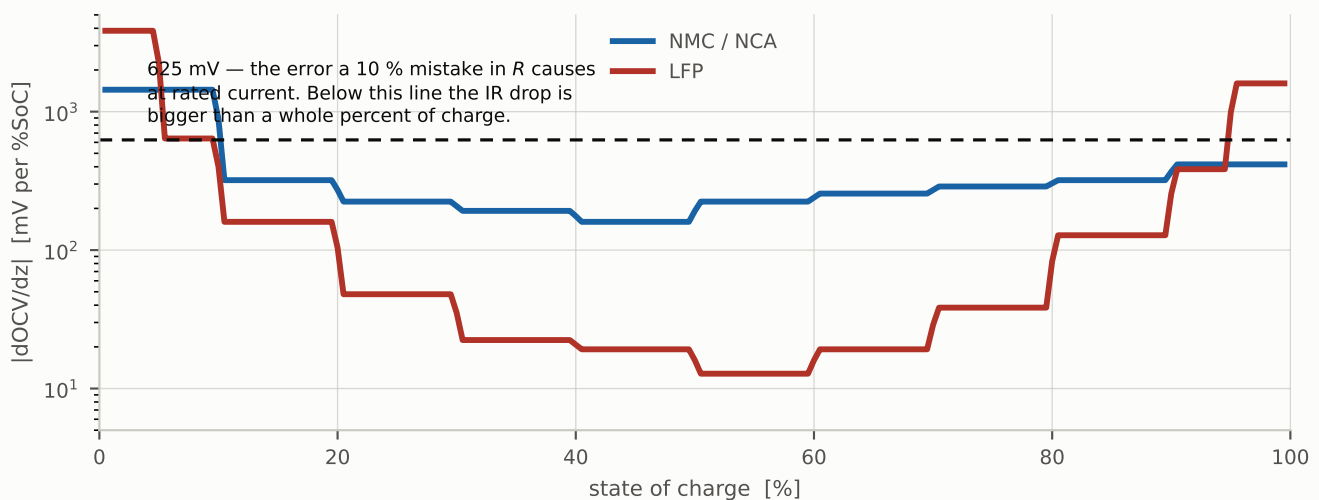
$$H = \frac{\partial V}{\partial z} = \frac{dOCV}{dz}$$

and the Kalman gain carries it:

$$K = \frac{PH}{H^2P + R_v} \implies H \rightarrow 0 \Rightarrow K \rightarrow 0$$

A flat curve means the filter learns nothing from a measurement, no matter how good the voltmeter is. This is not a tuning problem. It is the chemistry.

Pack-level slope, log scale. LFP's plateau sits an order of magnitude below NMC's across the entire working range.



The same filter, two chemistries

Measured from this project's own constants — 32S pack, 417 A rated, 15 mΩ, and a 10 % uncertainty in R (a 625 mV error at rated current):

	slope, mid-range	SoC error
NMC / NCA	208 mV/%SoC	3.0 %
LFP	18 mV/%SoC	34 %

An eleven-fold difference, from the curve alone. The literature reports the same shape of result — roughly 8 % against 49 % — and LFP estimators are known to diverge for exactly this reason.

Sources: Plett, G. L. (2004), 'Extended Kalman filtering for battery management systems of LiPB-based HEV battery packs', Parts 1-3, J. Power Sources 134(2):262-276. Slope figures cross-checked against published NMC and LFP OCV characterisations. Regenerate this sheet with docs/make_estimator_formulas.py.